

Digital Asset Settlement for Zelle

Executive Architecture Brief for Stablecoin and Cross-Border Settlement

Author	John Whitton
Version	1.0.1
Publication date	16 July 2026
Estimated reading time	18 minutes
Audience	Zelle hiring manager and Principal Engineer interview panel Engineering directors, senior architects, and technical executives

Recommendation. Extend the participant-bank model with an invisible, evidence-gated settlement rail. Keep the internal ledger authoritative, Java/Spring in control, chain-specific technology behind adapters, and the first pilot to one asset, one network, one corridor, no consumer wallet, and no bridge.

Evidence and scope notice. Zelle is used only as a public case study. This brief does not describe or infer confidential Early Warning Services architecture, vendors, controls, data, service levels, production plans, or technology selections. It visibly distinguishes public facts, source-backed inference, proposed architecture, organization-profile recommendations, assumptions, and unknowns.

Abstract

This brief proposes how a regulated digital-asset settlement platform could extend a bank-distributed payment experience without turning the customer journey into a crypto product. Zelle's publicly described model—a recognizable service delivered through participating financial-institution applications—makes it a useful case study for separating customer experience from settlement infrastructure. The architecture retains bank channels, aliases, fraud controls, compliance, support, and disputes; adds a Java/Spring control plane, authoritative double-entry ledger, durable payment intent, treasury, controlled signing, network adapters, independent observation, and reconciliation; and treats blockchain as one external leg rather than the source of business truth.

The immediate recommendation is deliberately narrow: discover the corridor and legal authorities, simulate the ledger and state machine, then prove one regulated asset on one approved public network for a limited participant cohort. The architecture does not select Ethereum, Base, Solana, a custody provider, or an issuer on Zelle's behalf. Each remains an evidence-gated decision. The result is a route that can be stopped, reconciled, and exited before expansion—and an engineering foundation that can later support additional corridors, assets, or networks without changing the customer proposition or the core financial invariants.

Contents

Abstract	2
1. The problem: a faster leg is not a complete payment	4
2. The opportunity for Zelle	4
3. Proposed architecture: control the payment, contain the rail	5
4. Ten decisions that keep the platform governable	6
5. Chain evaluation: choose the route, not the brand	9
6. Recommended pilot: a bounded route with a real exit	10
7. Wallet and custody recommendation: design roles before providers	11
8. Java and the engineering boundary	12
9. Roadmap: expand evidence before scope	13
10. Principal risks and executive control posture	14
11. First 90 days: Principal Engineer plan	15
12. Relationship to the engineering series	16
A. Public claim to primary source map	18
Annotated Primary Public Sources	19

Executive thesis. Preserve the bank-owned customer experience and regulated control plane; use a digital asset only as a bounded, reconciled settlement leg. Prove one route before expanding any dimension.

1. The problem: a faster leg is not a complete payment

PUBLIC FACT The global cross-border agenda still targets four persistent outcomes: lower cost, greater speed, better transparency, and broader access. In official remarks, the FSB Deputy Secretary General points to poor data quality and limited standardization as processing frictions, while World Bank corridor data shows that consumer remittance cost remains material on average. The remarks expressly represent the speaker's views rather than necessarily those of the FSB or its members. (Moloney 2026; World Bank 2025) Those are end-to-end problems. They are not solved merely by making a token transfer fast.

A conventional cross-border payment can pass through customer authentication, sanctions and fraud decisions, funding, foreign exchange, correspondent or payout networks, recipient validation, liquidity, accounting, exceptions, support, and legal discharge. Each step has a different authority, clock, failure mode, and repair mechanism. Delay is often the accumulated result of prefunding, cutoffs, time zones, fragmented data, manual exception handling, and uncertainty about who bears a loss. A new settlement leg can improve some of those mechanics, but it can also introduce issuer, chain, custody, privacy, and provider risks.

INFERENCE A regulated stablecoin can offer a programmable, continuously available transfer instrument on a shared network. It may reduce the time between controlled release and recipient-side availability, make a transaction independently observable, and support more direct treasury movement. It does not establish that the sender was entitled to pay, the recipient may lawfully receive, reserves are available, a bank account was credited, a customer promise was fulfilled, or an obligation was legally discharged. FATF's current work also emphasizes that stablecoin reach and interoperability create real illicit-finance and cross-chain control obligations. (Financial Action Task Force 2026)

Architecture thesis. Blockchain can make one external settlement event more direct and observable. The regulated platform must still own intent, policy, accounting, finality judgments, customer status, reconciliation, repair, and loss allocation.

The most dangerous design shortcut is therefore a single state called "settled." A chain can finalize while a payout fails. A customer can see completion while accounting remains open. An accounting period can close while legal ownership is disputed. A provider can report success while an independent node disagrees. Collapsing these conditions makes happy paths look simple by moving ambiguity into incidents, support queues, and financial breaks.

The opportunity is not to replace the payment network with a blockchain. It is to add an explicitly bounded settlement capability behind the network's business controls. The success measure is not transactions per second in isolation. It is whether a defined corridor can complete with correct balances, known authority, governed keys, explicit finality, recoverable ambiguity, timely reconciliation, and a credible exit path at an acceptable cost.

The decision question

Can a participant-facing route produce better end-to-end cost, time, transparency, or liquidity outcomes while preserving—and proving—the controls expected of a regulated payment network? If the answer cannot be demonstrated for one bounded corridor, expansion is premature.

2. The opportunity for Zelle

PUBLIC FACT Zelle publicly describes a network that connects financial institutions and is experienced through the mobile banking applications of participant banks and credit unions. Its current partner materials report availability through more than 2,400 banking and credit-union applications and more than 143 million consumers; the page identifies its institutional survey as Q1 2026. (Zelle 2026) These facts establish distribution and participant context; they do not establish internal architecture, vendors, or a cross-border plan.

That distribution model is strategically important because a settlement innovation need not require a new consumer destination. A bank can remain the customer channel and account relationship. Existing alias patterns, authentication, fraud controls, customer communication, support, and dispute processes can remain at the experience boundary. A digital asset can operate behind that boundary between approved institutional roles, provided the end-to-end obligation and customer treatment are explicit.

PROFILE DECISION The proposed Zelle-profile posture is therefore participant-facing and blockchain-invisible. A customer initiates a familiar business payment through a participating bank. The network creates a durable payment intent, applies policy, reserves liquidity, and coordinates a settlement route. Participants and operators receive business status, not wallet seed phrases, gas

prompts, chain addresses, or arbitrary contract access. The recipient receives value through an approved payout or account-credit mechanism. Exceptions remain in governed support and operations workflows.

This creates three forms of optionality:

- **Product optionality.** The customer proposition can remain stable while the network evaluates a new rail. A failed pilot does not strand customers with tokens or keys.
- **Network optionality.** The enterprise contract can support Ethereum, Base, Solana, or another approved network behind adapters without making one chain's SDK the platform domain model.
- **Strategic optionality.** Evidence from actual route economics, control performance, and operational failure can inform later decisions about additional assets, networks, issuer relationships, or an organization-specific L2.

The participant model also clarifies who must own each decision. Banks own customer authentication and account treatment under the agreed scheme. The settlement platform owns intent, orchestration, ledger, policy coordination, participant status, and reconciliation. Treasury owns liquidity and asset exposure. Security and custody controls own signing authority. The selected network owns its consensus rules. Issuers, payout partners, and providers own bounded external responsibilities. Legal agreements define discharge and loss allocation. No one system inherits every authority simply because it has the fastest clock.

What the case study does not assert

The public facts do not show that Zelle currently uses a blockchain, has selected Solana or another network, issues a token, operates a custody system, or plans a specific corridor. Those remain explicit unknowns and profile decision slots. The brief answers what should be built and validated if the business chooses to explore the opportunity.

3. Proposed architecture: control the payment, contain the rail

PROPOSED The architecture divides responsibility into four layers: customer and participant channels; the regulated control plane; bounded settlement capabilities; and independent evidence, reconciliation, and operations. This is a proposed target architecture derived from Volume I, not a claim about an existing Zelle system.

Evidence status: **PROPOSED** proposed architecture. Component names and responsibilities derive from Volume I and accepted ADRs; no box represents a confirmed EWS deployment.

3.1. Enterprise control plane

Participant APIs accept approved business requests such as create payment, reserve liquidity, release settlement, and query status. Every request becomes a durable payment intent before an external side effect. Policy coordination binds identity, sanctions/fraud disposition, limits, asset, network, corridor, signer policy, and approvals to that intent. The state machine records operations and immutable attempt lineage. A transactional outbox and durable consumer deduplication aim for exactly-once business effects without claiming impossible universal exactly-once delivery.

PROPOSED The internal double-entry ledger remains authoritative for obligations, reservations, positions, fees, suspense, corrections, and close. A chain receipt, custodian statement, issuer report, bank file, or payout confirmation is independent evidence to reconcile—never a replacement for the complete platform record. Corrections are linked reversals or adjustments, not destructive edits. See [ADR-001](#) and [Volume I: Ledger and State](#).

3.2. Bounded settlement capabilities

Treasury and liquidity services allocate prefunded positions and gas/fee capacity. Custody or an HSM/MPC signing boundary evaluates a narrowly scoped signing request; applications never receive raw production keys. Chain adapters translate a business operation into native transaction semantics while retaining nonce, blockhash, fee, contract/program, and attempt evidence. Issuer, FX, bank, and payout integrations use the same bounded gateway pattern.

Managed infrastructure may submit transactions, but a materially independent read path observes inclusion, canonicity, token state, and relevant events. Submission success is not observation truth. A timeout after possible acceptance becomes an ambiguous outcome; the platform investigates the original identity rather than creating a new value-moving intent.

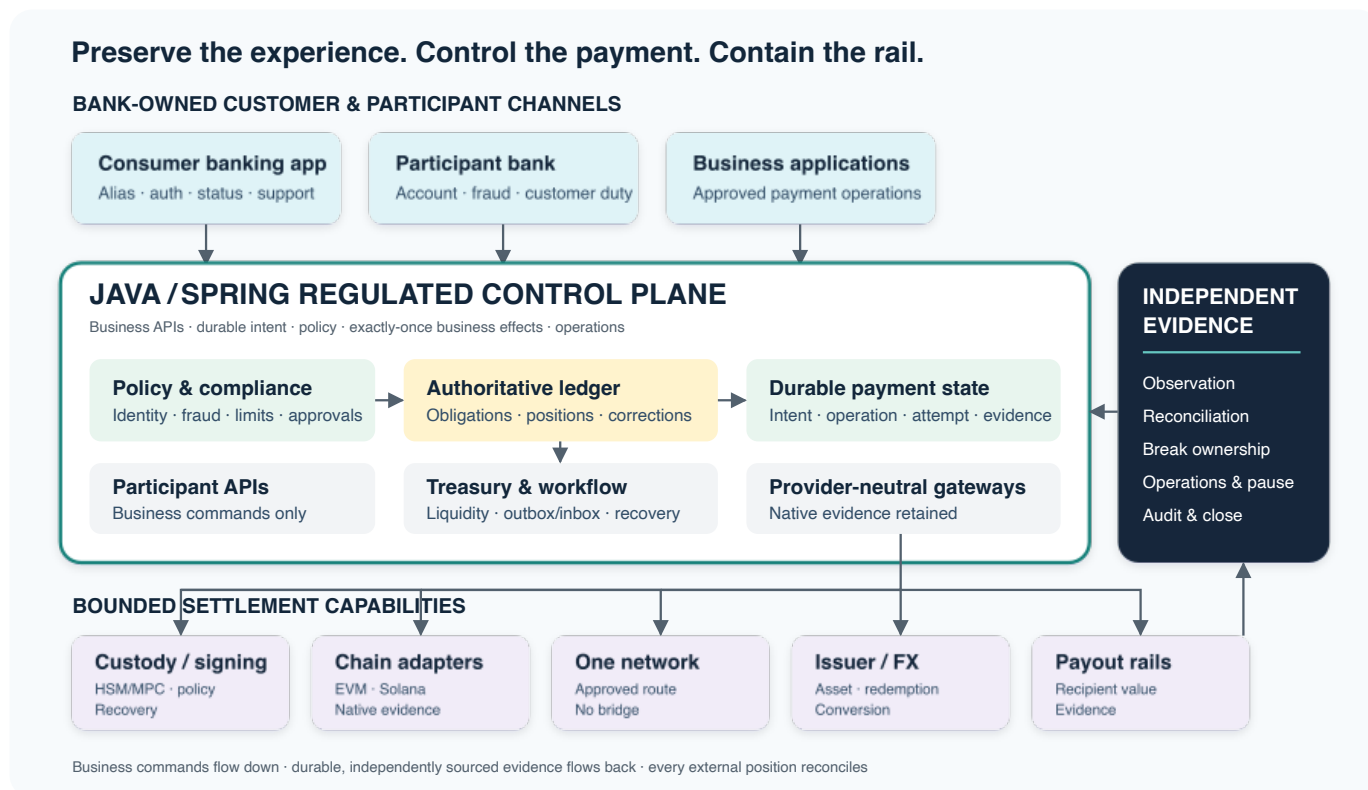


Figure 1: Executive ecosystem architecture. Bank-owned channels invoke business operations in a Java/Spring control plane. The internal ledger, durable state, policy, and reconciliation remain authoritative while custody, networks, issuers, FX, and payout rails remain bounded external systems.

3.3. Evidence, reconciliation, and operations

Reconciliation compares ledger and workflow records with chain data, custody balances, issuer statements, bank accounts, FX trades, fees, and payout outcomes. Every break has identity, owner, severity, age, evidence, disposition, and governed repair. Operations receives pause, replay, investigate, compensate, rotate, and exit capabilities at business boundaries rather than unrestricted database or RPC access.

The architecture is intentionally asymmetric. The regulated platform owns business truth and recovery. External systems own only their bounded native facts. Every boundary preserves enough evidence to explain disagreement.

4. Ten decisions that keep the platform governable

These decisions are the executive control surface. Each recommendation points to a canonical ADR; the brief summarizes its business consequence without creating a second authority. Profile-specific items remain proposed until their named gate closes.

01 · The internal ledger remains authoritative

PROPOSED

Recommendation. Record every obligation, reservation, fee, position, suspense item, and correction in a controlled double-entry ledger. Treat all external records as reconciliation evidence.

Why. Only the platform ledger spans the complete payment and its repair history. **Tradeoff.** The platform owns financial correctness, close, replay, and repair complexity.

Alternatives rejected. Chain state, a provider database, or a bank statement as platform truth. **Residual risk.** Posting defects and incomplete reconciliation.

Gate. Balance, precision, concurrency, replay, correction, restore, and close tests with financial review. **ADR-001.**

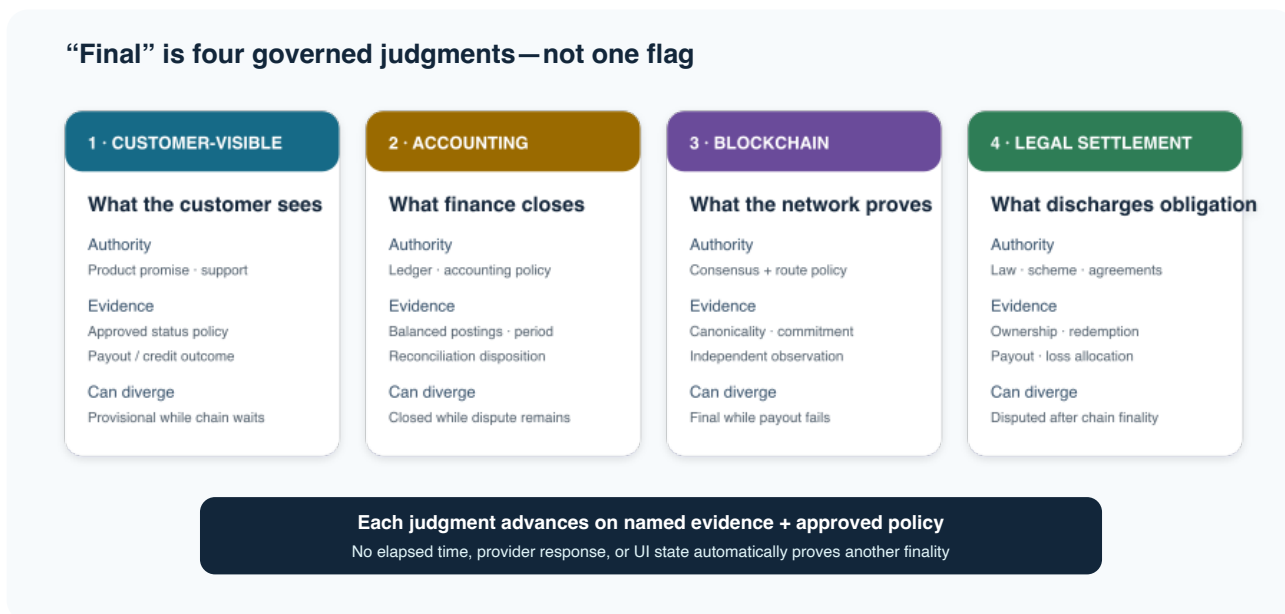


Figure 2: Four independent finalities. Each judgment has its own authority, evidence, customer or operational consequence, and correction path.

02 · Blockchain is an external settlement leg

PROPOSED

Recommendation. Submit only approved business operations through chain adapters; retain native transaction and observation evidence.

Why. A chain can establish its own state, not customer entitlement, bank payout, accounting close, or legal discharge.

Tradeoff. Adapters and reconciliation add engineering work.

Alternatives rejected. Chain as the payment system of record or an unrestricted privileged RPC gateway. **Residual risk.** Providers or observers can disagree.

Gate. Independent observation, ambiguous-outcome recovery, provider-failure drill, and exit test. **ADR-013.**

03 · Four finalities remain independent

PROPOSED

Recommendation. Model customer-visible completion, accounting finality, blockchain finality, and legal settlement finality as independent judgments with named evidence and policy.

Why. The four authorities can advance or diverge independently. **Tradeoff.** APIs, storage, support tooling, and operations become more explicit.

Alternatives rejected. One settled flag or a simple business/chain split. **Residual risk.** Users may still misinterpret a status.

Gate. Test every pairwise divergence; obtain accounting and legal approval for route-specific evidence. **ADR-002.**

Evidence status: **PROPOSED** canonical architecture decision from ADR-002. Economic finality is a chain-risk threshold inside blockchain policy, not a substitute for legal settlement finality.

04 · Customer blockchain complexity remains invisible

PROFILE DECISION

Recommendation. Preserve the bank-owned experience; do not require consumers to hold tokens, manage seed phrases, pay gas, or understand chain addresses in the initial pilot.

Why. The pilot should isolate settlement learning from a new custody, recovery, disclosure, and support product. **Tradeoff.** The platform retains backend responsibility.

Alternatives rejected. Consumer token or wallet at pilot launch. **Residual risk.** Invisible failures can still affect customers.

Gate. Journey, disclosure, support, dispute, and loss-allocation review. **ADR-003; SLOT-WALLET-001.**

05 · Java/Spring owns enterprise orchestration**PROFILE DECISION**

Recommendation. Use Java/Spring for participant APIs, durable intent, policy, ledger integration, messaging, reconciliation, observability, and operations.

Why. These are long-lived enterprise controls with transactional and operational demands. **Tradeoff.** The platform remains polyglot at native chain boundaries.

Alternatives rejected. Moving enterprise authority into chain-native services or forcing contracts/programs into Java.

Residual risk. Schema and adapter drift.

Gate. Contract tests, threat model, observability, load test, and failure drills. **ADR-004**; SLOT-EXEC-001.

06 · Chain technology stays behind adapters**PROPOSED**

Recommendation. Keep EVM JSON-RPC, Solana commitment/blockhash rules, custody SDKs, and provider types behind business-level contracts.

Why. Intent, ledger, idempotency, and finality invariants should survive a chain or provider change. **Tradeoff.** A lowest-common-denominator abstraction would be unsafe. Native evidence must remain available.

Alternatives rejected. Vendor SDK types in the domain or an arbitrary contract-call API. **Residual risk.** Lossy abstraction and privileged escape paths.

Gate. Native dossier, policy allowlists, conformance tests, and replacement adapter. **ADR-005**.

07 · Start with one asset, one network, one corridor**PROFILE DECISION**

Recommendation. Limit participants, notional, duration, asset, network, corridor, and operating windows; prefund liquidity and expand one dimension at a time.

Why. A narrow route produces interpretable evidence and a credible rollback. **Tradeoff.** Learning is intentionally incomplete and prefunding consumes capital.

Alternatives rejected. Multi-asset/network launch or a demo that never exercises participant operations. **Residual risk.** Results may not generalize.

Gate. Entry criteria, success measures, stop conditions, exit rehearsal, and independent approval. **ADR-006**; SLOT-ROADMAP-001.

08 · No bridge by default**PROFILE DECISION**

Recommendation. Keep the initial route on one native network with no generic bridge.

Why. A bridge adds contracts or programs, validators or attestors, liquidity, replay, finality composition, governance, incident, and exit risk. **Tradeoff.** Reach and network flexibility are constrained.

Alternatives rejected. A generic cross-chain route in the pilot; issuer-native interoperability remains a later evaluation.

Residual risk. Single-network concentration.

Gate. Automated topology scan and a separate architecture/security/treasury/legal decision for any exception. **ADR-015**.

09 · Qualified custody or HSM/MPC controls signing**PROFILE DECISION**

Recommendation. Bind signing to policy, transaction identity, allowlists, quorum, dual control, recovery, audit, and provider exit. Never place production keys in application memory.

Why. Cryptographic authority is a financial control, not an SDK convenience. **Tradeoff.** Integration, ceremony, and diligence cost increase.

Alternatives rejected. Application keys, direct consumer signing, or custody labels without control analysis. **Residual risk.** Authentication, recovery, and concentration can defeat nominal MPC.

Gate. Compromise, pause, rotation, recovery, export, and in-flight migration drills. **ADR-012**; SLOT-CUSTODY-001.

10 · L2 or issued stablecoin only after evidence**PROFILE DECISION**

Recommendation. Do not build a proprietary L2 or issue an organization-specific stablecoin for the pilot. Reopen either choice only when measured need justifies the new obligations.

Why. An L2 introduces sequencer, settlement, data-availability, bridge, upgrade, gas, governance, and exit duties. Issuance introduces reserve, redemption, legal, compliance, and loss-allocation duties. **Tradeoff.** Near-term strategic control is

lower.

Alternatives rejected. Build now for hypothetical scale or permanently prohibit future evaluation. **Residual risk.** Strategic pressure can bypass gates.

Gate. Separate business case and ADR with legal, security, treasury, operations, recovery, and exit proof. **ADR-010;** SLOT-L2-001.

5. Chain evaluation: choose the route, not the brand

VALIDATE No public evidence establishes a Zelle chain selection. The correct executive decision is therefore a gated comparison tied to the pilot asset, corridor, participants, legal rights, liquidity, custody support, privacy posture, failure tolerance, and exit plan. A universal chain winner would be less rigorous than a route-specific decision.

Evidence status: **PUBLIC FACT** narrow protocol facts refreshed 16 July 2026 (Ethereum Foundation 2026; Base 2026a; Base 2026b; Solana Foundation 2026c; Solana Foundation 2026a);

PROPOSED qualitative working hypotheses; **PROFILE DECISION** candidate disposition.

Table 1: Decision summary; the complete dimension record is in data/chain-comparison.csv.

Candidate	Cost posture	Finality posture	Engineering fit	Principal operating risk	Disposition
Ethereum	Variable L1 gas; benchmark	Native PoS justified/finalized states	Mature EVM/Solidity; direct JSON-RPC/Web3j	Fee/capacity variability and public governance	Eligible baseline; not selected
Base	Lower execution target plus L1 data/security fee	L2 inclusion then L1 publication/finality; withdrawals differ	EVM compatible; direct JSON-RPC/Web3j	Sequencer, L2 governance, and L1 dependency	Eligible candidate; not selected
Solana	Low base-fee design plus optional priority fee	Processed, confirmed, and finalized commitments	Rust/Anchor native; neutral service boundary for Java	Commitment/RPC/account contention and native skills	Eligible candidate; not selected
Future Zelle L2	Unknown until workload and stack design	Must compose sequencer, settlement, DA, and exit	Could be EVM-compatible; full platform must be built	Highest governance, bridge, security, and exit burden	Not applicable to pilot

5.1. What the public sources establish

PUBLIC FACT Ethereum documents transaction progression through inclusion, justification, and finalization. (Ethereum Foundation 2026) “Finalized” remains a consensus judgment, not legal discharge.

PUBLIC FACT Base documents a two-part fee model: L2 execution plus L1 security/data cost. Its finality documentation separates fast sequencer/L2 inclusion from L1 batch publication and batch finality, with distinct L2-to-L1 withdrawal behavior. (Base 2026a; Base 2026b)

PUBLIC FACT Solana exposes processed, confirmed, and finalized RPC commitments. Its fee model combines a base fee with an optional compute-unit priority fee. (Solana Foundation 2026c; Solana Foundation 2026a)

PROPOSED The remaining comparative judgments—including ecosystem maturity, throughput posture, custody support, operating burden, governance exposure, and strategic control—are decision hypotheses. They are included to expose diligence questions, not to establish current facts or a winner. Before selection they require authoritative refresh, route benchmarks, provider diligence, native-domain review, sensitivity analysis, and profile approval. Low nominal fees or headline throughput cannot replace failure drills and end-to-end route evidence.

PROFILE DECISION A future Zelle L2 is a decision slot, not a candidate for the minimum pilot. Strategic control must be weighed against the obligation to own or govern sequencing, settlement, data availability, upgrades, fees, infrastructure, monitoring, incident response, interoperability, and exit. “Control” alone is not a business case.

5.2. Promotion gates

Before selection, the decision team must demonstrate: the regulated asset is natively available and redeemable for the intended entities; legal and accounting treatment is approved; finality thresholds map to the four-finality model; peak and stress cost/latency are measured; custody and observation paths satisfy independence and recovery; privacy exposure is accepted; liquidity is available; and an alternate or termination path can close all obligations. The profile records the result under SLOT-CHAIN-001; the neutral architecture does not change.

Four candidates. One route-specific decision. No selection asserted.

	COST	FINALITY PATH	ECOSYSTEM / TOOLING	OPERATING BURDEN	ZELLE CONTROL
ETHEREUM Eligible · not selected	Variable L1 gas benchmark route	Native PoS justified → finalized	Deep EVM maturity Solidity · Web3j		Low neutral public rail
BASE Eligible · not selected	L2 + L1 fees lower target; variable	L2 → L1 batch withdrawals differ	EVM compatible OP Stack operations		Low–mid external sequencer
SOLANA Eligible · not selected	Low fee design priority fee optional	RPC commitments processed / confirmed / final	Rust / Anchor native distinct program model		Low neutral public rail
FUTURE ZELLE L2 Not applicable to pilot	Unknown full lifecycle cost	Must design sequencer · DA · settlement	Must build / operate ecosystem not asserted		Potentially high with highest duty

PROPOSED ROUTE HYPOTHESES — VALIDATE BEFORE SELECTION
 Asset · legal rights · liquidity · custody · privacy · finality · resilience · provider exit

Figure 3: Executive chain comparison. Narrow protocol facts inform proposed qualitative working hypotheses across cost, finality, tooling, operating, and control dimensions. The bands are not verified comparative facts, measured scores, or a selection.

6. Recommended pilot: a bounded route with a real exit

PROFILE DECISION The minimum credible pilot is one approved regulated stablecoin, one approved public network, one corridor, and a limited participant cohort. It should move enough controlled value to exercise treasury, signing, chain observation, payout, support, and accounting—but remain small enough to pause, reconcile, compensate, and terminate without systemic consequence.

Evidence status: **PROFILE DECISION** proposed Zelle-profile pilot. Asset, chain, corridor, participants, providers, and deployment remain validation gates.

6.1. Pilot envelope

Dimension	Proposed boundary
Asset	One regulated, redeemable stablecoin with approved issuer, reserve, freeze, redemption, and legal treatment. PUBLIC FACT USDC is an example of an issued multi-network asset, not a selection. (Circle 2026)
Network	One native public network selected through SLOT-CHAIN-001; no generic bridge and no cross-network in-flight obligation.
Corridor and cohort	One approved send/receive corridor; named participant banks or controlled internal entities; explicit customer, notional, velocity, and duration limits.
Liquidity	Prefunded treasury or omnibus positions as an operating assumption, with ownership, accounting, limits, rebalancing, and emergency exit approved before value moves.
Custody and infrastructure	Qualified custody or HSM/MPC; managed write submission paired with a materially independent observation path. No provider is preselected.
Customer boundary	No direct consumer wallet, token balance, gas, bridge, seed phrase, or chain-address workflow.
On-chain logic	Use issuer-standard transfer behavior; no custom contract or program unless an irreducible shared rule is demonstrated and separately reviewed.

One asset · one network · one corridor · one explainable lifecycle

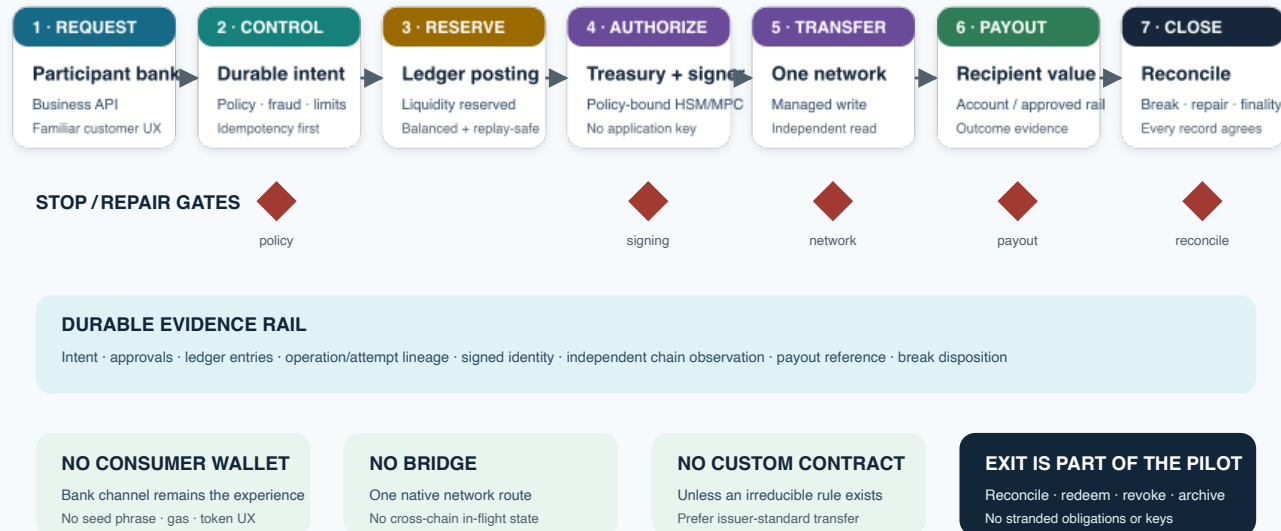


Figure 4: Bounded pilot flow. A durable intent and reserved ledger position precede compliance, signing, chain submission, independent observation, payout, and reconciliation. Every external step can pause or enter controlled repair.

6.2. Entry criteria

The pilot starts only after the legal entity and licensing map, asset rights, accounting treatment, participant agreements, customer treatment, sanctions/fraud model, custody authority, network policy, liquidity model, support playbook, and loss-allocation matrix are approved. The control plane must pass deterministic ledger, idempotency, ambiguity, reorg/finality, signing, reconciliation, restore, and pause tests. Production credentials and limits must be created through dual-control ceremonies.

6.3. Success, stop, and exit

Success is an evidence set, not a launch date: zero unbalanced postings or duplicate business effects; every intent and external attempt explainable; reconciliation breaks within approved aging; observed cost and latency within route thresholds; no policy bypass; recovery and provider failover demonstrated; support and participant outcomes accepted; and every finality judgment traceable to evidence and policy.

Stop new intents automatically or by governed operator action when legal approval changes, reserve/redemption confidence falls, the asset deviates beyond limits, liquidity buffers breach, signing authority is suspect, independent observers materially disagree, the network or provider exceeds outage/finality thresholds, reconciliation breaks age beyond tolerance, sanctions/fraud controls degrade, or customer loss allocation becomes unclear. Stopping new work does not abandon in-flight obligations: the state machine moves them to controlled completion, return, compensation, or manual repair.

Exit means more than switching off an API. The team must export addresses, balances, attempts, policies, approvals, and audit evidence; reconcile ledger, custody, issuer, chain, bank, and payout positions to zero or an approved retained balance; redeem or transfer the asset; revoke and archive signing authority; complete customer/participant treatment; and preserve records. Only a rehearsed exit makes a pilot genuinely bounded.

7. Wallet and custody recommendation: design roles before providers

“Wallet” is too broad to be an architecture decision. The platform must first name who can initiate, approve, sign, recover, pause, and exit for each value role. Only then can it evaluate custody technology or providers.

Table 2: Wallet and authority roles for the bounded pilot.

Role	Authority and experience	Pilot posture
Consumer	Initiates through the bank experience; no direct token/key authority	No consumer wallet
Participant bank	Authenticates customer, applies bank controls, invokes business APIs	No raw signer or arbitrary RPC
Settlement/operator	Holds narrowly scoped rail authority for approved intents	Policy-bound institutional wallet
Issuer/mint	Issues, redeems, freezes, or administers the asset under issuer rules	External authority; reconcile and monitor
Treasury	Funds, rebalances, pays fees, and manages exposure/limits	Separate duties, limits, and addresses
Administration	Configures allowlists, contracts/programs, policies, pause, and upgrades	Stronger quorum and change ceremony
Recovery	Restores or migrates authority under exceptional conditions	Offline/independent ceremony and tested exit

PROFILE DECISION The minimum pilot should not require direct user wallets. A custodial versus non-custodial consumer-wallet choice would add authentication, device binding, transaction consent, recovery, account portability, privacy, sanctions, support, and legal-custody questions without improving the initial settlement-learning objective. If a future product demonstrates customer value, it should reopen SLOT-WALLET-001 as a separate product and architecture decision.

PROFILE DECISION Institutional settlement authority should use qualified custody or an HSM/MPC boundary with explicit policy. “MPC,” “non-custodial,” or “embedded” is not an adequate control description. The review must identify who authenticates, who holds shares or keys, who defines policy, whether a provider can act alone, how destination and amount allowlists bind to the signed payload, how recovery changes authority, and how the organization exits with complete evidence.

The signing request should carry the payment intent, operation and attempt identity, approved asset/network, contract or program, method/instruction, amount, destination, fee bounds, nonce or blockhash, expiry, policy version, and approvals. The signer returns a signature or transaction identity plus durable audit evidence. A lost response remains ambiguous until activity logs and network observation determine whether the original request was accepted.

Custody control test

Assume the application credential is stolen, one operator is malicious, the provider is unavailable, a recovery administrator is compromised, and the organization must migrate providers during in-flight payments. The design passes only if policy still prevents unauthorized value, legitimate obligations can be recovered, and evidence remains complete.

See **ADR-011** and **ADR-012**. Provider evaluation belongs in the reusable engineering reference library and a dated procurement/control review, not in the neutral architecture.

8. Java and the engineering boundary

PROFILE DECISION Java/Spring should own the regulated enterprise control plane: participant APIs, durable payment intent, authorization and policy coordination, ledger integration, transactional messaging, workflow, reconciliation, case/operations interfaces, telemetry, and controlled repair. This is a profile recommendation aligned with the architecture’s responsibilities, not a claim about EWS’s current stack.

PUBLIC FACT Spring’s official documentation covers web, messaging, transactional data/persistence, integration, scheduling, caching, observability, and testing concerns relevant to long-lived enterprise services. (Spring Team 2026) The architecture benefit is not the brand itself. It is the ability to keep business authority, transactional boundaries, policy, and operational evidence in a mature control-plane environment with clear ownership.

PUBLIC FACT Web3j describes itself as a Java library for Ethereum JSON-RPC and smart-contract integration. (Web3j 2026) It can encode transactions, call nodes, consume receipts/events, and generate typed wrappers. It does not implement Ethereum’s public consensus, establish legal finality, or make Java an EVM contract language. Production signing should be delegated to the approved custody/HSM boundary rather than implemented with application-held credentials.

Solidity remains the native language for EVM contracts. (Solidity Team 2026) The pilot should avoid a custom contract unless the issuer-standard transfer model cannot express an irreducible shared rule. If a contract is required, its scope, upgrade/admin authority, audit, monitoring, pause, and exit are separate security decisions.

Table 3: Use the language that owns each execution environment.

Boundary	Primary role	Why it belongs there
Java/Spring	Business API, intent, ledger, policy, workflow, messaging, reconciliation, operations	Owns regulated enterprise state and long-lived service contracts
Web3j / EVM client	Typed JSON-RPC, transaction, receipt, event, and contract-wrapper integration	Adapter library inside the Java boundary; not a chain or consensus engine
Solidity	Minimal EVM smart-contract rules when shared on-chain enforcement is irreducible	Native EVM execution and security semantics
Rust / Anchor	Solana programs and native services where the Solana model requires them	Native account, instruction, CPI, compute, and deployment semantics
TypeScript or Rust service	Optional native adapter when ecosystem support is materially stronger	Remains behind the same language-neutral domain contract

PUBLIC FACT Solana documents Rust as the primary language for programs compiled to its chain-native sBPF environment. (Solana Foundation 2026b) A serious Solana option therefore requires first-class Rust/Anchor, account/PDA, instruction/CPI, token, compute-budget, transaction-lifetime, RPC commitment, testing, deployment, and monitoring expertise. An all-Java mandate would either ignore native semantics or force an immature abstraction. The control plane can remain Java while a neutral adapter contract permits a native implementation.

Boundary rule. Java owns the payment; native technology owns the chain interaction; the adapter contract ensures neither can silently take authority from the other.

9. Roadmap: expand evidence before scope

The roadmap treats every phase as an investment option with explicit promotion and stop gates. It does not commit to production scale before the preceding phase proves financial correctness, authority, recoverability, and an end-to-end business outcome.

Evidence status: **PROFILE DECISION** proposed implementation roadmap under SLOT-ROADMAP-001; dates and organization commitments remain unknown.

- Discovery and control definition.** Select candidate corridor and participants; map legal entities, obligations, systems of record, customer treatment, data, risks, and authority boundaries. Exit if the business benefit or legal path is not credible.
- Ledger, state, and simulation.** Implement the exact-money ledger contract, durable intent, attempt lineage, outbox/inbox, four finalities, and reconciliation using synthetic data. Simulate duplicates, reordering, timeouts, reorgs, provider disagreement, payout failure, corrections, and restore.
- Bounded internal pilot.** Integrate sandbox or controlled external systems, institutional signing, managed submission, independent reads, treasury, and operations. Exercise the complete stop and exit process before customer or participant exposure.
- Participant API pilot.** Add a limited participant cohort behind business APIs, contractual limits, customer/support treatment, production controls, and high-touch monitoring. Expand only after success measures remain stable through normal and adverse conditions.
- Cross-border production.** Promote one corridor with approved service levels, capacity, on-call ownership, reconciliation close, compliance/fraud governance, financial controls, and continuity. Retain explicit route termination and provider exit.
- Multi-asset or network expansion.** Add one dimension at a time. Each asset, network, corridor, participant class, custody model, or provider receives an impact review and route-specific evidence. Cross-network movement is a new architecture decision.
- Evidence-based L2 or issuer decision.** Only measured limitations can justify a proprietary L2 or organization-issued stablecoin. Evaluate the full lifecycle, not just transaction cost or control. Deferral or rejection is a successful outcome when evidence does not justify the obligation.

Portfolio rule

Never expand asset, network, corridor, participant population, custody authority, and notional limit in the same release. Changing one dimension preserves the ability to attribute outcomes and roll back safely.

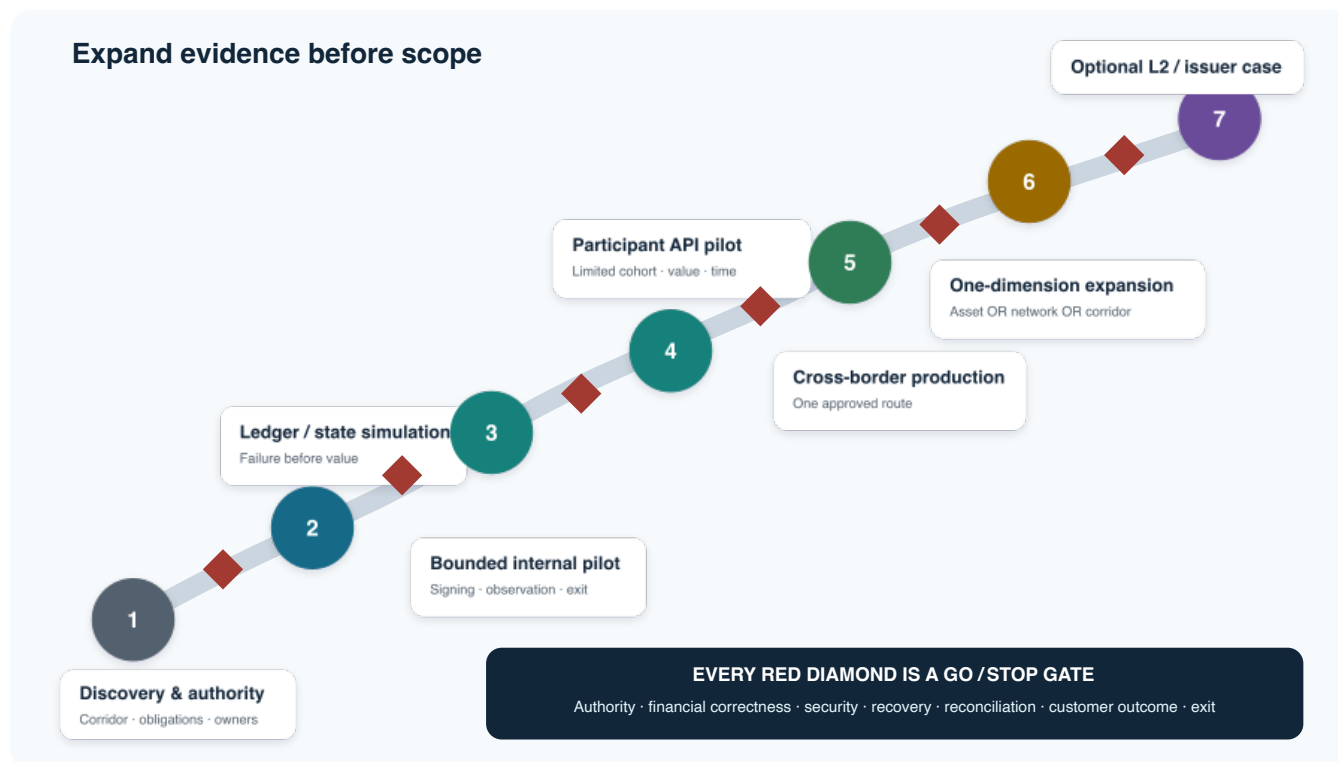


Figure 5: Evidence-gated roadmap. Discovery and simulation precede bounded value; participant production precedes multi-asset or multi-network expansion; an L2 or issuer decision is last and optional.

10. Principal risks and executive control posture

Risk is not a separate appendix to the architecture. It determines the authority model, state machine, pilot limits, provider boundaries, operating evidence, and stop conditions. The register below summarizes the minimum executive posture; data/risks.csv preserves stable IDs, owners, residual risk, and validation gates.

Table 4. Principal risk ownership, controls, and residual exposure.

ID	Risk	Control posture	Residual / gate
001	Regulatory and licensing	Map entities, activities, corridor, asset, counterparties, disclosures, and legal finality before design approval.	High until qualified written disposition; no value before approval.
002	Issuer and reserve	Diligence reserve/redemption/freeze rights; concentration limits; redemption and alternate-exit tests.	Issuer dependency remains; de-peg/freeze scenarios must pass.
003	Chain availability and finality	Network-specific policy, independent reads, queuing, pause, reorg and governance playbooks.	Public-network risk remains; chaos and contradiction drills must pass.
004	Custody and key authority	HSM/MPC or qualified custody, dual control, payload policy, allowlists, ceremonies, recovery, and exit.	Concentrated authority remains; compromise/rotation/export rehearsals must pass.
005	Sanctions and fraud	Identity, screening, transaction monitoring, velocity/amount controls, holds, and case management.	Adaptive adversaries and false positives remain; model and case drills required.
006	Liquidity	Forecasts, prefunding limits, buffers, alerts, rebalancing, fees, and governed stop thresholds.	Capital cost and stress risk remain; emergency rebalance must pass.

ID	Risk	Control posture	Residual / gate
007	Privacy	Minimize on-chain/provider data; address strategy, access, retention, and correlation-risk review.	Public metadata cannot be eliminated; data-flow approval required.
008	Provider concentration	Neutral contracts, independent observation, evidence export, replacement integration, and termination assistance.	Specialist dependencies remain; failover and migration must pass.
009	Smart-contract	Prefer issuer-standard transfer; minimize custom logic; verify code/admin authority; audit, allowlist, monitor, pause.	Token contract risk remains; exact deployed artifact review required.
010	Bridge	No bridge in the pilot; separate threat model and ADR for any exception.	Low in pilot only if topology confirms absence.
011	Reconciliation	Independent records, first-class breaks, aging/ownership, immutable correction, daily close, and restore.	Late/conflicting data remains; repair drills and aging SLO required.
012	Customer loss allocation	Explicit status, agreements, support, disputes, reversal/compensation rules, and scenario matrix.	High until every partial failure has approved treatment.

PUBLIC FACT FATF's current stablecoin work highlights illicit-finance risks involving stablecoins, unhosted wallets, and cross-chain activity. (Financial Action Task Force 2026) The response is not to assume that blockchain analytics substitutes for regulated controls. Identity, sanctions, fraud, transaction context, issuer authority, wallet control, chain evidence, payout, and customer treatment must work as an end-to-end decision system with reviewable model governance.

10.1. Failure containment priorities

Prevent duplicate value. Durable idempotency, uniqueness, attempt lineage, outbox/inbox, and reconciliation enforce one permitted economic effect even when delivery repeats. A timeout after possible submission is not permission to create a fresh transfer.

Separate pause from repair. A pause blocks new intents or a bounded operation class. It does not mutate history or erase obligations. In-flight work moves to observation, controlled completion, return, compensation, or manual repair with dual control.

Preserve raw evidence. Provider-normalized success is insufficient for audit and recovery. Retain request identity, signed payload or digest, network transaction identity, observation source and policy, block/slot context, issuer/custody records, payout references, and operator decisions.

Design exit before adoption. Provider or chain exit requires portable addresses, policy and audit history, replacement signing/submission/observation, in-flight handling, asset redemption or transfer, and a ledger-to-external close. A nominally non-custodial service can still be operationally non-portable.

Executive stop principle

When authority, evidence, or loss allocation is unclear, stop creating new obligations. Availability targets never authorize the platform to guess whether value moved.

11. First 90 days: Principal Engineer plan

PROFILE DECISION The first ninety days should reduce irreversible uncertainty before moving value. The deliverable is a jointly owned set of authority maps, decision records, executable contracts, simulations, and pilot gates—not a premature chain integration.

Table 5: First-90-day outcomes.

Window	Focus	Evidence produced
Days 1–30	Authority and current-state discovery	System-of-record map; participant/customer journey; legal-entity and obligation map; data and trust boundaries; current controls; decision and unknown register; candidate corridor and participant criteria
Days 31–60	Contracts and simulation	Exact-money ledger model; durable intent/state/finality contracts; API and event schemas; signing policy; reconciliation model; synthetic happy and failure paths; chain/custody evaluation scorecards
Days 61–90	Pilot readiness and operating proof	Candidate route dossier; measured sandbox behavior; ambiguity/re-org/provider/custody/restore drills; support and loss-allocation scenarios; limits and stop conditions; phased plan and go/no-go review

11.1. 1. Validate architecture and authority boundaries

Interview product, participant, payment operations, fraud, compliance, legal, treasury, finance, security, SRE, support, and platform owners. Trace one representative payment and every exception across systems. Name who can authorize, observe, reverse, compensate, pause, recover, and declare each form of finality. Convert conflicts into explicit decisions rather than smoothing them into a diagram.

11.2. 2. Define systems of record and canonical decisions

Confirm the ledger authority and chart-of-account implications; identify current customer, participant, fraud, case, treasury, and settlement records; and establish a single ADR for each material choice. Revalidate the profile rather than assuming Phase 0 recommendations are organization decisions. Record chain, asset, custody, infrastructure, deployment, corridor, and participant selections as open gates.

11.3. 3. Build financial and failure simulation first

Implement executable ledger postings and state transitions with synthetic data before any production signing. Simulate duplicate requests, redelivery, out-of-order events, lost responses, partial approval, network congestion, reorg or commitment disagreement, issuer freeze, signer outage, payout failure after chain finality, liquidity depletion, period close, correction, backup restore, and provider exit. Every scenario must preserve balance, lineage, evidence, and a governed next action.

11.4. 4. Select the pilot corridor and controls

Score candidate corridors on customer value, participant readiness, legal clarity, currency and payout access, stablecoin rights, liquidity, fraud/sanctions exposure, privacy, operational support, and ability to exit. Then evaluate asset, chain, custody, and providers for that route. The ordering matters: selecting a chain before the obligation and corridor would optimize the wrong system.

Day-90 decision

Proceed only if the route has a credible business benefit, approved authority, balanced and replay-safe accounting, explicit four-finality policy, controlled signing, independently observable network evidence, complete reconciliation, rehearsed stop/exit, and named owners. Otherwise continue discovery or stop without treating schedule as failure.

12. Relationship to the engineering series

This brief is the first organization-profile edition of a vendor-neutral publication contract. It exports a decision narrative, pilot posture, risk frame, and first-90-day roadmap for an executive audience. It does not become the authority for ledger semantics, finality, detailed chain engineering, or implementation.

Series artifact	Relationship	Authority
Normative kernel	Owns evidence taxonomy, invariants, publication/profile contracts, governance, and release gates	Level 0
Volume I: Reference Architecture	Canonical layered architecture, ledger/state/finality model, flows, security, reconciliation, technology boundary, pilot, and roadmap	Architecture
Canonical ADR library	Owns each decision once; this brief links rather than duplicates authority	Decision
PROFILE-ZELLE	Applies registered chain, wallet, custody, SDK, infrastructure, roadmap, L2, and executive-tailoring slots	Configuration
Future Volume II	Will own detailed engineering patterns, native chain dossiers, provider evaluations, implementation guidance, tests, deployment, and runbooks	Specification only
Future Volume III	Will illustrate the neutral architecture in runnable unbranded code	Not begun

The deeper path for a reviewer is deliberate: start here for the recommendation; use Volume I to challenge the architecture; use the ADRs to examine alternatives and consequences; use the Zelle profile to see which values are organization-specific; and later use Volume II for implementation depth. The claim ledger in this release preserves the exact public, inferred, proposed, profile-specific, assumed, and unknown boundaries behind the executive wording.

What I would build first if hired. Not a token and not a bridge: an executable ledger and payment-intent model, a failure simulator, and reconciliation that can prove what happened. Those controls make every later chain, custody, and corridor choice safer and faster.

Release scope. This editorial release is limited to the Executive Brief. Volume II and Volume III have not begun. All organization choices remain subject to qualified internal review and the registered profile gates.

A. Public claim to primary source map

This appendix maps the brief's major factual topics to primary public sources. It is intentionally concise rather than sentence-by-sentence. Normal citations remain beside public facts in the narrative; the internal claim ledger retains the complete identifiers, evidence classes, source locations, caveats, and review dispositions used for regression and release validation.

Table 6. Major public claims, source coverage, and evidence limits.

Topic	Primary source(s)	Supports	Limitations
Zelle network	Zelle and Early Warning Services (Zelle 2026; Early Warning Services 2016)	Participant-bank distribution, current public scale figures, Q1 2026 survey date, and public launch context.	Does not establish internal architecture, vendors, technology, or a stablecoin and cross-border plan.
Cross-border payments	FSB remarks and World Bank data (Moloney 2026; World Bank 2025)	Cost, speed, transparency, access, data-quality, standardization, and remittance-price context.	Does not prove a Zelle-specific problem or that the proposed route improves end-to-end outcomes.
Ethereum	Ethereum Foundation documentation (Ethereum Foundation 2026)	Inclusion, justification, and finalization terminology.	Does not establish legal, accounting, or customer finality; route performance; or selection.
Base	Base documentation (Base 2026a; Base 2026b)	L2 execution and L1 security/data fee components; L2 inclusion, L1 publication/finality, and withdrawal stages.	Does not establish fixed cost, service levels, comparative superiority, or selection.
Solana	Solana Foundation documentation (Solana Foundation 2026c; Solana Foundation 2026a; Solana Foundation 2026b)	RPC commitment levels, base and priority fees, and Rust/sBPF program execution.	Does not establish route performance, legal finality, operating fitness, custody support, or selection.
Spring	Spring Framework documentation (Spring Team 2026)	Framework capability areas relevant to enterprise services.	Does not establish EWS adoption, production reliability, or suitability without engineering validation.
Web3j	Web3j documentation (Web3j 2026)	Java integration with Ethereum JSON-RPC and smart contracts.	Does not provide consensus, legal finality, production key controls, or evidence of organization adoption.
USDC	Circle issuer documentation (Circle 2026)	Circle's description of a multi-network issued asset and its backing statement.	Is not independent reserve assurance and does not select an asset for the pilot.
FATF guidance	Financial Action Task Force report (Financial Action Task Force 2026)	High-level illicit-finance concerns involving stablecoins, unhosted wallets, and cross-chain activity.	Does not decide a pilot's legal treatment or prove that any control stack is sufficient.
Solidity	Solidity documentation (Solidity Team 2026)	Solidity's role in EVM smart-contract development.	Does not establish that custom contracts are needed, safe, or the right place for enterprise control.

The annotated source list that follows supplies the organization, title, publication date, canonical URL, supported use, and explicit limitation for every primary source used by this release.

Annotated Primary Public Sources

Base (2026a). *Network Fees*. URL: <https://docs.base.org/base-chain/network-information/network-fees> (visited on 2026-07-15).

Supports: The Base fee model's L2 execution component and L1 security or data component.

Does not establish: A fixed transaction price, total end-to-end operating cost, comparative cost superiority, or chain selection.

– (2026b). *Transaction Finality*. URL: <https://docs.base.org/base-chain/network-information/transaction-finality> (visited on 2026-07-15).

Supports: Distinct Base stages for L2 inclusion, L1 batch inclusion and finality, and L2-to-L1 withdrawal behavior.

Does not establish: A service-level guarantee, legal finality, route suitability, comparative performance, or chain selection.

Circle (2026). *What is USDC?* URL: <https://developers.circle.com/stablecoins/what-is-usdc> (visited on 2026-07-15).

Supports: Circle's description of USDC as a multi-network issued stablecoin and its statement about liquid cash and cash-equivalent backing.

Does not establish: Independent reserve assurance, legal or accounting treatment, route availability, or selection of USDC for the pilot.

Early Warning Services (Oct. 24, 2016). *Early Warning Announces Zelle Network*. URL: <https://www.zellepay.com/press-releases/early-warning-announces-zelle-network> (visited on 2026-07-15).

Supports: The public 2016 announcement of the Zelle name and its bank-led distribution context.

Does not establish: Current network scale, internal implementation, present strategy, or any stablecoin and cross-border architecture.

Ethereum Foundation (Mar. 12, 2026). *Transactions*. URL: <https://ethereum.org/developers/docs/transactions/> (visited on 2026-07-15).

Supports: Ethereum transaction progression and the protocol terms inclusion, justification, and finalization.

Does not establish: Legal, accounting, or customer finality; route latency; a service-level guarantee; or chain selection.

Financial Action Task Force (2026). *Targeted Report on Stablecoins and Unhosted Wallets*. URL: <https://www.fatf-gafi.org/en/publications/Virtualassets/targeted-report-stablecoins-unhosted-wallets.html> (visited on 2026-07-15).

Supports: High-level illicit-finance concerns involving stablecoins, unhosted wallets, and cross-chain activity.

Does not establish: A legal disposition for a specific pilot, the sufficiency of a control stack, or a conclusion about Zelle.

Moloney, M. (July 8, 2026). *Cross-Border Payments: Towards the Next Chapter*. URL: <https://www.fsb.org/2026/07/cross-border-payments-towards-the-next-chapter/> (visited on 2026-07-15).

Supports: The four cross-border outcomes of cost, speed, transparency, and access, plus processing friction from poor data quality and limited standardization.

Does not establish: An institutional FSB position, a Zelle-specific problem statement, or proof that a stablecoin route improves an end-to-end payment.

Solana Foundation (2026a). *Fees on Solana*. URL: <https://solana.com/docs/core/fees> (visited on 2026-07-15).

Supports: The Solana base-fee design and optional compute-unit-based prioritization fee.

Does not establish: A fixed route cost, comparative cost superiority, congestion behavior, or chain selection.

– (2026b). *Program Execution*. URL: <https://solana.com/docs/core/programs/program-execution> (visited on 2026-07-15).

Supports: Rust as the primary Solana program language and compilation to the chain-native sBPF execution environment.

Does not establish: The enterprise control-plane language, program safety, implementation fitness, or chain selection.

– (2026c). *Solana RPC Methods and Documentation*. URL: <https://solana.com/docs/rpc> (visited on 2026-07-15).

Supports: The processed, confirmed, and finalized commitment levels exposed by Solana RPC.

Does not establish: Legal or customer finality, route-specific economic safety, RPC service levels, or chain selection.

Solidity Team (2026). *Solidity Documentation*. URL: <https://docs.soliditylang.org/en/latest/> (visited on 2026-07-15).

Supports: Solidity's role as a language for EVM smart-contract development.

Does not establish: The need for custom contracts, contract safety, enterprise orchestration ownership, or a chain decision.

Spring Team (2026). *Spring Framework Documentation*. URL: <https://docs.spring.io/spring-framework/reference/index.html> (visited on 2026-07-15).

Supports: Spring Framework capability areas relevant to web, messaging, transactions, persistence, integration, scheduling, caching, observability, and testing.

Does not establish: EWS or Zelle adoption, system reliability, architectural fitness without evaluation, or production performance.

Web3j (2026). *Web3j Documentation*. URL: <https://docs.web3j.io/latest/> (visited on 2026-07-15).

Supports: Web3j's role as a Java library for Ethereum JSON-RPC and smart-contract integration.

Does not establish: Consensus implementation, legal finality, secure production signing, version fitness, or organization adoption.

World Bank (2025). *Remittance Prices Worldwide*. URL: <https://remittanceprices.worldbank.org/> (visited on 2026-07-15).

Supports: The continued materiality of average consumer remittance cost across the corridors monitored by the World Bank database.

Does not establish: The price of every corridor, a current Zelle route, or the cost benefit of the proposed pilot.

Zelle (2026). *Innovation. Powered by Partnerships*. URL: <https://www.zelle.com/join-zelle-network/partners> (visited on 2026-07-16).

Supports: The participant financial-institution model, availability through more than 2,400 banking and credit-union applications, more than 143 million consumers, and the Q1 2026 survey date.

Does not establish: Internal architecture, vendors, ledger implementation, technology stack, or a cross-border and stablecoin plan.